

- 7 A light elastic band of length 20 cm is stretched to a length of 28 cm, and then allowed to return back to its original length. The graph in Fig. 7.1 below shows how the length of the band varies with the force applied.

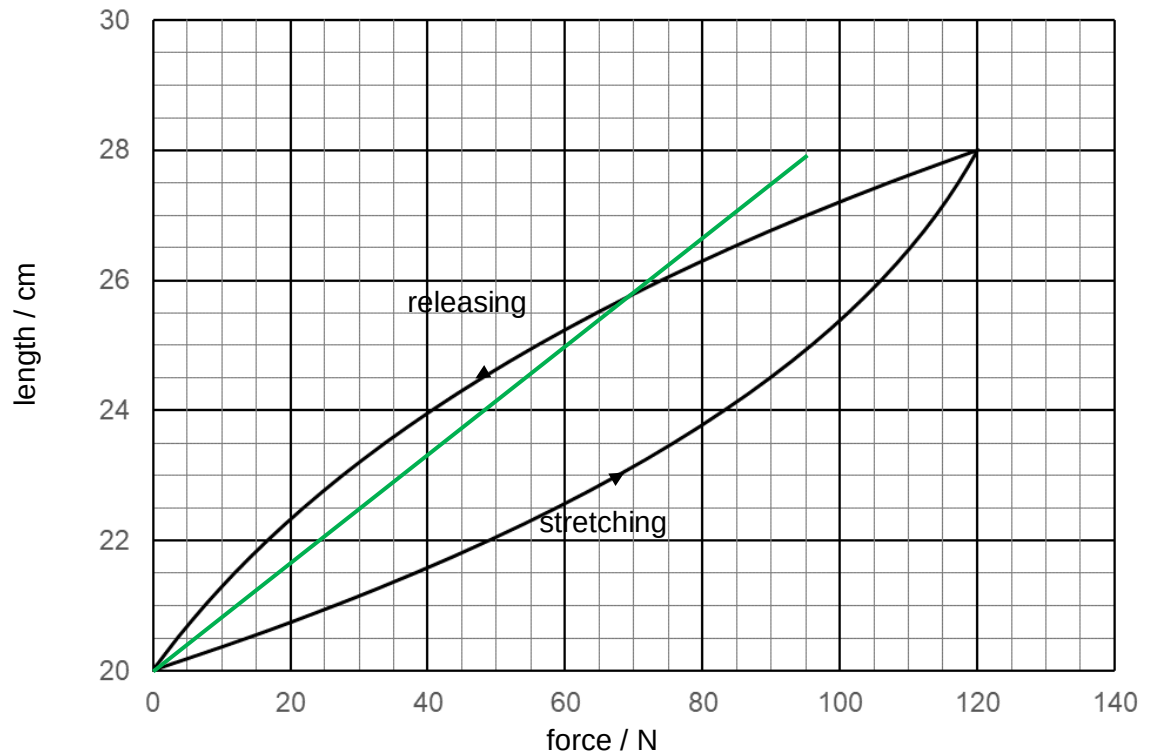


Fig. 7.1

- (a) Explain how the graph shows that the band does not obey Hooke's Law.

The graph is not a straight line/does not have a constant gradient [1]

This shows that force applied is not proportional to extension [1]. However, this proportionality relation is required of the band if it were to obey Hooke's law. [2]

- (b) Estimate the work done by the applied force to stretch the band.

Work done = Area between stretching graph and Length-axis [1]

$$= 15 \text{ squares} \times (0.02 \text{ m} \times 20 \text{ N})$$

$$= 6.0 \text{ J } (\pm 0.3 \text{ J}) [1]$$

work done = _____ J [2]

- (c) The elastic band is used in a catapult to project a pellet into the air. The band is stretched to a length of 28 cm and released. Explain using Fig. 7.1, why the kinetic energy of the pellet at the point of release is significantly smaller than your answer in (b)

Area between length axis and release curve is smaller than between the length axis and stretch curve [1]

This implies that only some of the stored elastic potential energy in the stretched band has been converted to kinetic energy of the pellet [1]

.....[2]

- (d) The initial kinetic energy of the projected pellet is 3.8 J. Calculate the efficiency of the band in accelerating the pellet.

efficiency = kinetic energy of pellet / initial EPE stored in band

= kinetic energy of pellet / WD in stretching band

$$= 3.8 / 6.0 = 0.63$$

efficiency = W [2]

- (e) Suggest one reason for the loss in efficiency.

Heat is produced when band is released.

.....[1]

- (f) The elastic band is replaced with another new elastic band of the same length which obeys Hooke's Law. It is also stretched to a length of 28 cm and transfers 3.8 J of kinetic energy to the pellet.

- (i) Calculate the spring constant of this band.

$$E = \frac{1}{2} k x^2$$

$$3.8 = \frac{1}{2} k \times (0.08)^2 \quad [1]$$

$$k = 1.2 \times 10^3 \text{ N m}^{-1} \quad [1]$$

spring constant = N m⁻¹ [2]

- (ii) Sketch the graph for this new elastic band on Fig 7.1. [1]

(Side working to determine final coordinate point for graph plotting:

At length = 28 cm, i.e. extension = 8 cm,

$$E = \frac{1}{2} F x$$

$$3.8 = \frac{1}{2} F \times 0.08 \rightarrow F = 95 \text{ N}$$

∴ Graph plotted should be a straight line starting from (0, 20) and ends at (95, 28)

[Total: 12]

8 The variation with time t of the displacement y of a wave A as it passes a point P is